

Activity- 5

AIM:

To observe the refraction and lateral deviation of a beam of light incident obliquely on a glass slab.

APPARATUS

Drawing board, rectangular glass slab, white sheet of paper, adhesive tape, drawing pins, a metre scale, Alpins, protractor, sharp pencil, eraser.

THEORY:

When a ray of light is incident on a rectangular glass slab, it is refracted through it. It emerges out of the slab parallel to the direction of the incident ray. The emergent ray suffers only a lateral displacement. For a given angle of incidence and a pair of media, the lateral deviation is proportional to the thickness of the glass slab.

PROCEDURE:

1. Fix a white paper sheet by drawing pins on a drawing board.
2. Take a glass slab and put it symmetrically in the middle of the paper, and mark its boundary ABCD.
3. Draw a normal at point Q on face AB and draw a line PQ making an angle i with the normal. PQ will represent an incident ray.
4. Fix two pins at points 1 and 2 on the line PQ at distances 1cm or more between themselves.
5. See images of these pins through the face DC, and fix two more pins at points 3 and 4 (1cm or more apart) such that these two pins cover the images of the first two pins, all being along a straight line.
6. Remove the glass slab. Draw a straight line RS through points 3 and 4 to represent the emergent ray
7. Join the QR to represent the refracted ray
8. Draw a normal at point R on face DC and measure angle e . It comes to be equal to angle i . Produce PQ forward to cut DC at T. Draw N perpendicular to RS. N measures lateral displacement (d)

RESULT:

- i. Angle of incidence(i)=angle of emergence(e)
- ii. The lateral displacement increases with the angle of incidence.

Precautions

- i. The angle of incidence should lie between 30° and 60°

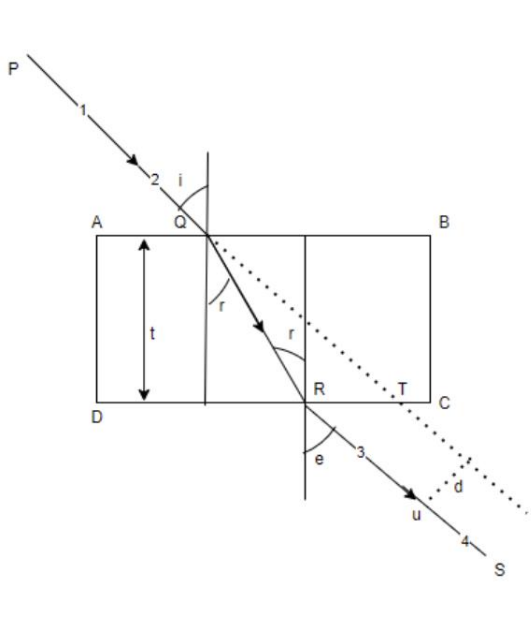
- ii. While fixing the pins P and Q or the pins R and S, care should be taken to maintain a distance of about 5cm between the two pins

Sources of error

- i. All pins' bases should lie in a straight line
- ii. All measurements of angles using a protractor should be done accurately

Diagram and observation table on the left side of the page

Diagram:



Observation table

S. No.	Angle of incidence	Angle of emergence	Lateral displacement
1			
2			
3			

